

BAYANBULAG, Mongolia

WMO: 442750

Lat: 46.8122N

Lon: 98.0869E

Elev: 2255

StdP: 77.01

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.7	-33.3	-38.2	0.1	-35.0	-36.0	0.2	-32.4	11.6	-16.1	9.8	-15.5	0.5	50	0.787

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	24.3	14.2	22.6	13.2	21.0	12.4	16.0	22.7	14.6	21.1	13.4	19.5	4.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.2	12.5	20.3	11.9	11.4	18.6	10.6	10.5	17.3	53.6	22.8	49.2	21.2	45.3	19.5	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	9.7	8.6	-38.2	27.5	2.6	1.7	-40.1	28.8	-41.6	29.8	-43.1	30.8	-45.0	32.0		
			-38.1	17.8	2.6	2.9	-40.0	19.9	-41.5	21.6	-43.0	23.2	-44.9	25.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.1	-21.8	-18.2	-10.5	-0.6	5.6	11.8	14.2	11.7	4.9	-3.4	-12.5	-19.2	(6)
(7)		DBStd	13.28	4.81	5.46	5.79	4.92	4.32	3.19	2.94	2.94	3.97	4.74	5.66	4.84	(7)
(8)		HDD10.0	5065	986	791	635	319	146	18	3	16	157	414	674	906	(8)
(9)		HDD18.3	7827	1244	1024	893	568	396	198	133	205	403	673	924	1164	(9)
(10)		CDD10.0	285	0	0	0	1	9	71	132	69	4	0	0	0	(10)
(11)		CDD18.3	4	0	0	0	0	0	0	4	1	0	0	0	0	(11)
(12)		CDH23.3	91	0	0	0	0	1	17	63	9	1	0	0	0	(12)
(13)		CDH26.7	6	0	0	0	0	0	1	5	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.6	1.4	1.8	2.9	3.8	3.9	3.1	2.7	2.5	2.6	2.5	2.1	1.6	(14)
(15)	Precipitation	PrecAvg	151	1	2	5	5	9	29	51	30	12	4	2	2	(15)
(16)		PrecMax	276	5	9	58	23	28	136	146	91	39	26	8	6	(16)
(17)		PrecMin	58	0	0	0	0	0	1	6	2	0	0	0	0	(17)
(18)		PrecStd	51	1	2	11	6	10	26	36	19	11	5	2	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.1	-0.6	8.3	16.2	22.2	25.4	27.5	24.6	20.4	13.2	6.0	-3.2	(19)
(20)			MCWB	-7.7	-4.3	1.9	8.2	11.0	14.2	16.9	14.8	10.4	5.9	0.9	-5.7	(20)
(21)		2%	DB	-8.1	-3.6	5.2	13.2	19.4	23.2	25.1	22.7	17.8	10.3	2.3	-6.1	(21)
(22)			MCWB	-10.0	-6.5	-0.5	5.5	10.1	13.0	15.1	13.4	9.2	4.2	-1.6	-8.3	(22)
(23)		5%	DB	-10.7	-5.8	2.6	10.9	17.0	21.4	23.3	21.0	15.7	8.1	-0.5	-8.3	(23)
(24)			MCWB	-12.2	-8.1	-2.0	4.1	8.7	12.4	13.9	12.6	8.2	2.3	-3.5	-10.1	(24)
(25)		10%	DB	-13.1	-8.4	-0.1	8.6	14.8	19.7	21.5	19.3	13.7	5.7	-3.3	-10.8	(25)
(26)			MCWB	-14.4	-10.2	-3.7	2.6	7.1	11.2	12.9	11.7	7.1	1.0	-5.7	-12.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.4	-4.1	2.6	9.0	13.1	16.2	18.7	16.5	12.0	6.8	1.1	-5.6	(27)
(28)			MCDWB	-5.6	-0.9	7.6	14.6	20.6	23.9	25.9	23.3	18.4	12.5	5.6	-3.4	(28)
(29)		2%	WB	-10.0	-6.4	-0.1	6.5	11.1	14.6	16.8	14.8	10.3	4.5	-1.3	-8.1	(29)
(30)			MCDWB	-8.2	-3.9	4.4	12.1	17.8	21.6	23.2	21.5	16.3	9.6	2.2	-6.2	(30)
(31)		5%	WB	-12.2	-8.1	-1.9	4.5	9.2	13.4	15.4	13.5	8.9	2.8	-3.4	-10.2	(31)
(32)			MCDWB	-10.7	-5.9	2.4	10.0	15.9	20.0	21.7	19.4	14.7	7.4	-0.5	-8.5	(32)
(33)		10%	WB	-14.3	-10.1	-3.7	2.8	7.6	12.1	14.0	12.3	7.4	1.1	-5.7	-12.1	(33)
(34)			MCDWB	-13.2	-8.4	-0.1	8.1	13.8	18.3	20.0	17.7	12.8	5.4	-3.4	-10.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.9	16.2	15.5	13.1	13.2	12.7	12.2	12.9	14.6	14.6	14.2	13.6	(35)
(36)			MCDBR	16.9	18.2	17.5	16.5	17.3	16.0	15.4	16.0	17.8	18.0	17.1	15.3	(36)
(37)			MCWBR	15.1	15.7	13.1	11.1	10.8	9.5	8.3	9.0	11.2	12.8	13.7	13.3	(37)
(38)		5% WB	MCDBR	16.6	18.1	17.1	16.0	16.0	14.8	14.0	14.4	17.0	17.5	16.8	15.0	(38)
(39)			MCWBR	15.0	15.8	13.2	11.8	11.2	9.8	8.9	9.2	11.7	13.1	13.6	13.3	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.216	0.244	0.285	0.324	0.334	0.344	0.342	0.319	0.286	0.263	0.236	0.218	(40)
(41)		taud		2.402	2.336	2.254	2.235	2.261	2.320	2.366	2.409	2.456	2.443	2.414	2.387	(41)
(42)		Ebn at Noon		919	946	942	930	929	918	915	927	935	910	878	870	(42)
(43)		Edh at Noon		66	89	114	129	130	124	117	107	92	79	65	60	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.95	3.05	4.57	5.97	6.95	6.88	6.53	5.86	4.97	3.39	2.09	1.58	(44)
(45)		RadStd		0.15	0.15	0.20	0.22	0.42	0.36	0.34	0.26	0.21	0.19	0.15	0.10	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	-1.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	-2.21	N/A	N/A	-1.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page